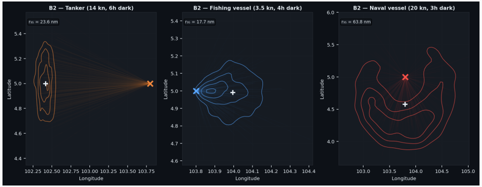
The Problem: Vessels Go Dark and Disguise Their Activity

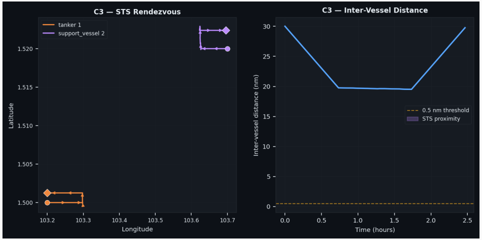
90,000 vessels of regulatory concern are active at any given moment — and the picture is incomplete by design.

75% of industrial fishing vessels intentionally disable their AIS transponders to avoid detection.

Even when a vessel *is* visible, analysts must answer two questions from limited, noisy data:

- **What is it doing?**
  Fishing · Transiting · Loitering · Ship-to-Ship transfer





Top: position uncertainty after AIS loss. Bottom: ship-to-ship transfer — a key evasion tactic.

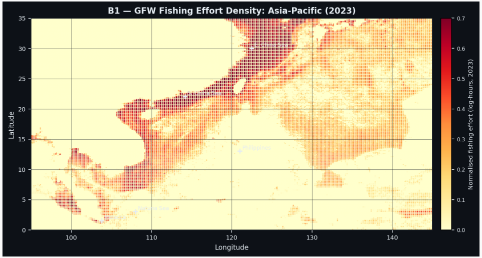
The Scale: Where Fishing Actually Happens

The system is grounded in **real fleet data** from Global Fishing Watch — 2023 effort across four operationally critical regions.

Region	Records	Fishing hrs
Brazil EEZ	309,711	1,382,042
Philippine EEZ	61,460	—
Strait of Malacca	54,040	—
Gulf of Guinea	59,954	—

Fishing effort is **not uniformly distributed** — it clusters around productive grounds and shifts seasonally.

Our synthetic simulator is calibrated against



GFW 2023 fishing effort — Asia-Pacific. Dense clusters mark productive grounds in the Philippine EEZ and South China Sea. The Strait of Malacca corridor is a key chokepoint for both fishing and transit traffic.

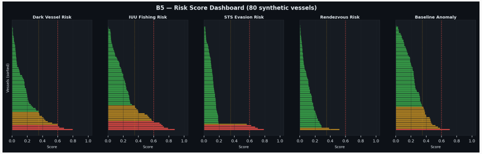
The System: Classify Any Vessel from Any Observation

We built an **Activity Intelligence Engine** that works regardless of how much data is available.

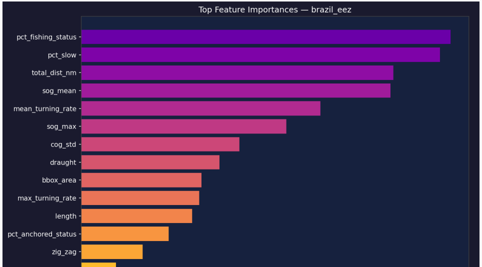
Inputs: AIS, radar, satellite imagery (EO/SAR), or any combination — down to a single observation.

Outputs per vessel:

- Activity class (fishing / transit / anchored / STS / loitering)
- Vessel type (12 classes)
- Dark-vessel risk score (0–1)
- Calibrated confidence — conservative

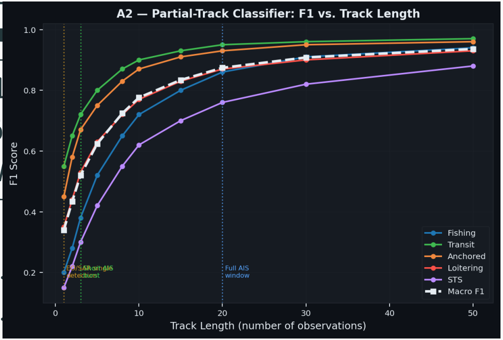


Risk score dashboard — IUU and dark vessel scores per vessel. All flagged vessels are fishing-registered.



Results: Robust Performance Across Regions and Sensors

What we measure	Resu
Activity classification F_1	**0.900–1**
Vessel type F_1 (12 classes)	**0.96**
Performance at $N = 1$ observation	**same as /**
Regions validated (synthetic)	4
Brazil EEZ	$F_1 = 0.$
Strait of Malacca	$F_1 = 0.$
Gulf of Guinea	$F_1 = 0.992$
Philippine EEZ	$F_1 = 0.992$
Dark vessels flagged	**49**
All fishing-registered	✓
Training time (216k examples)	**82 seco**



Classification accuracy is flat from a single observation to a full 50-ping track — no minimum custody duration needed.